

Solution Requirements

Date: 15 March 2026

Team ID: SWTID-2026-7387

Project Name: FitBuddy

Maximum Marks: 4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform. Functional requirements define specific behaviors, functions, or features of the system (what the system should do). Non-functional requirements specify the criteria that judge the operation of a system rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can register and securely log in using email/mobile credentials. Passwords are securely stored and users can reset forgotten passwords.	High
2	Authorization levels	FitBuddy supports user and administrator roles. Users manage their own profile, fitness goals and activity data; administrators manage system content and users.	High
3	External interfaces	The system can integrate with device/health APIs where available to collect steps, workouts and calories, and may use notification services.	Medium
4	Transactions processing	FitBuddy records workout completion, activity entries, goal progress and other fitness transactions consistently, with validation before saving data.	High
5	Reporting	The system generates dashboards and progress reports showing workouts, activity, goals, calories and progress over daily, weekly and monthly periods.	High
6	Business rules, etc.	The system provides personalized workout recommendations based on user goals and recorded activity. Goals, workout completion and progress calculations follow defined fitness rules.	High
7	Compliance to laws or regulations	The system protects personal and fitness-related data using appropriate privacy, consent, access-control and secure data-handling practices applicable to the deployment region.	High
8	Other	Users can create/update profiles, set fitness goals, view recommended workouts, record activities, track progress and receive reminders or notifications.	High

FitBuddy Functional Workflow

The functional workflow below expands the requirements into the main user-facing operations of the FitBuddy solution.

Area	Details
User Profile	Register/login, enter profile information, select fitness goals and maintain preferences.
Fitness Planning	Set goals such as activity, workout frequency and progress targets; receive suitable workout suggestions.
Workout Tracking	Start/complete workouts, record activities and maintain a history of completed sessions.
Progress Monitoring	Calculate and display progress using activity, workout and goal data through charts or summaries.
Notifications	Send reminders for planned workouts and progress-related notifications based on user preferences.
Administration	Authorized administrators can manage users, workout content, categories and system-level information.

Step 2: Non-Functional Requirements (NFR)

The FitBuddy system should meet the following operational quality targets and acceptance criteria.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages/screens and normal user actions should respond quickly; dashboard calculations should not cause noticeable delay.	Typical page/API response ≤ 2 seconds under normal load; common actions complete within 3 seconds.
2	Scalability	The architecture should support increasing users, workout records and activity history without major redesign.	Initial target: support at least 10,000 registered users and 1,000 concurrent users.
3	Security & Data Privacy	Protect account credentials and personal/fitness data using secure authentication, authorization, encrypted transport and controlled data access.	HTTPS/TLS for data in transit; passwords hashed securely; role-based access; no unauthorized access to user data.
4	Reliability & Availability	The service should remain available and preserve user data during routine failures or connectivity interruptions.	Target availability $\geq 99.5\%$ per month; validated data writes and regular backups.
5	Usability & Accessibility	The interface should be simple, mobile-friendly and understandable for users with different levels of technical familiarity.	Core tasks should be completed with clear navigation; readable text and accessible controls; responsive on common mobile/desktop screens.
6	Maintainability & Portability	Use modular components, documented APIs and maintainable code so features can be updated and deployed across supported environments.	Documented modules/APIs; version-controlled code; changes to one module should minimize impact on unrelated features.

Project Summary: FitBuddy is a fitness-support application designed to help users manage fitness goals, track workouts and activity, monitor progress, and receive personalized guidance and reminders through a secure and user-friendly system.